

StudyBuddy Wireframes

Seven mid-fi mobile wireframes covering an end-to-end study journey: *tomorrow's class card* → *briefing* → *equation picker* → *sourced worked solution* → *committed plan* → *wrap-up*, with a paper-capture flow alongside. Each screen traces to coded findings from the four interviews (P1–P4) and to a numbered requirement. State shown is illustrative; the design does not assume a fixed study rhythm (see §2.1 for the full rationale).

AUTHOR	Anker Rasmussen
MODULE	IN3065, User-Centred System Design
DELIVERABLE	Coursework Part 2 · §2 Wireframes (40%)
METHOD	Reflexive thematic analysis (Braun & Clarke 2006) → POV → requirements → wireframes
FORMAT	Mobile · 312 × 720 dp · Inter · single accent (Figma blue)

PRIMARY JOURNEY



SEVEN SCREENS

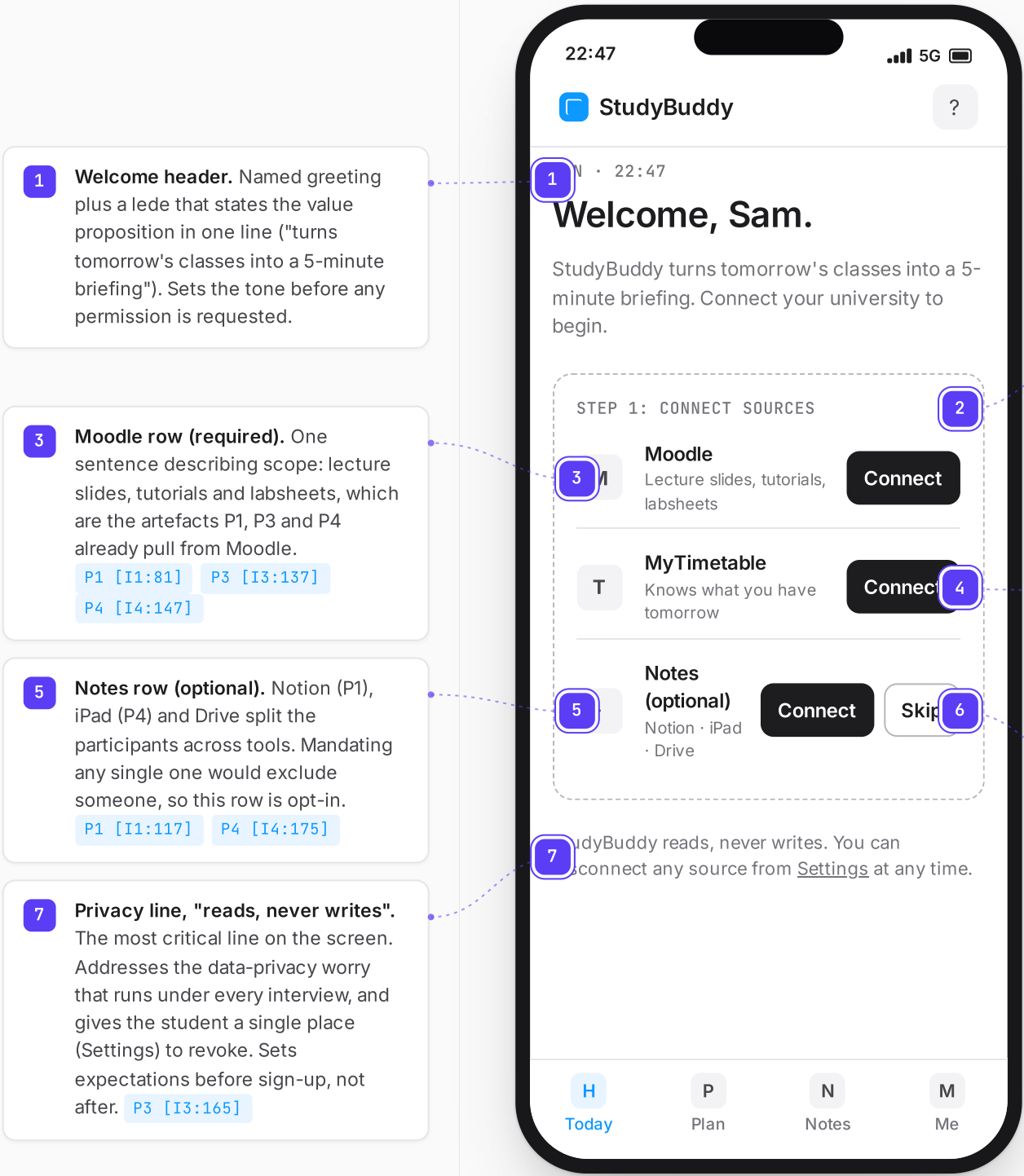
01	Home: tomorrow's briefing card	02	Briefing detail: recap + prep
03	Equation picker: "which one?"	04	Step-by-step worked solution
05	Session plan / commit	06	Wrap-up · felt readiness
07	Camera capture: paper input		

THEMES HIT PER SCREEN

T1	"Which equation?" and worked examples.	Screens 02, 03, 04, 07. P1, P2, P3, P4
T2	Personal prioritisation, rarely tracked.	Screens 01, 05, 06. P1, P2, P3, P4
T5	Moodle as channel; external memory beside it.	Screens 01, 02, 04. P1, P2, P3, P4
T3	Focus engineered.	Annotated only; covered through plan and last-session refresher.
T4	Group study.	Out of scope. The brief permits an individual-only design.

READING THE WIREFRAMES

Each frame pairs the rendered screen (centre) with numbered spec-card callouts (flanking) and a full annotation panel (right). Callouts explain **behaviour**, **states and source attribution**, not visual description. Citations: [P2](#) [\[I2:77\]](#) = participant and transcript line. [R3](#) = requirement ID. [Norman · 3 models](#) = HCI principle (course).



1 **Welcome header.** Named greeting plus a lede that states the value proposition in one line ("turns tomorrow's classes into a 5-minute briefing"). Sets the tone before any permission is requested.

3 **Moodle row (required).** One sentence describing scope: lecture slides, tutorials and labsheets, which are the artefacts P1, P3 and P4 already pull from Moodle.
P1 [I1:81] P3 [I3:137]
P4 [I4:147]

5 **Notes row (optional).** Notion (P1), iPad (P4) and Drive split the participants across tools. Mandating any single one would exclude someone, so this row is opt-in.
P1 [I1:117] P4 [I4:175]

7 **Privacy line, "reads, never writes".** The most critical line on the screen. Addresses the data-privacy worry that runs under every interview, and gives the student a single place (Settings) to revoke. Sets expectations before sign-up, not after. P3 [I3:165]

2 **"Step 1: Connect sources".** Numbered so the student knows this is scaffolded onboarding, not the final screen. No skip-all affordance: at least Moodle and a timetable are required for the rest of the app to work. Norman · system image

4 **MyTimetable row (required).** Sub-line states the value: "knows what you have tomorrow". The pre-class briefing cannot rank events without it. R5

6 **Connect or Skip, both visible.** The optional row exposes both options so the student can opt in *or* opt out in one tap, without going back. Skip is "Skip" — not "later" or "remind me" — so the choice is final for this run. Cooper · experience goal

Home: tomorrow's briefing card

The first-run state. Until the student connects at least a timetable and a slides source, no briefing can be synthesised, so the home screen becomes a single-purpose onboarding surface with an explicit data-scope line.

HCI PRINCIPLE APPLIED

Norman · 3 models: "Step 1" plus the explicit read-only line communicate the designer's intent ("connect a timetable, then we synthesise") in two lines, closing the gap between the designer's model and the student's before any credential is shared. **Cooper · experience goal:** optional Notes row has an explicit Skip; required rows do not.

EVIDENCE: INTERVIEW ANCHORS

Why an explicit privacy line, rather than a buried T&Cs link? P3's AI-cynicism P3 [I3:165] and P2's hallucination concern P2 [I2:169] both reduce to the same worry: *I don't trust where this data goes*. Surfacing read-only scope on the same screen as the connect buttons is the cheapest way to bound that anxiety before any credential is shared. The required-versus-optional split also mirrors the four participants' actual tool diversity: three of four use Moodle, only one uses Notion, only one an iPad.

REQUIREMENTS TRACED

R5

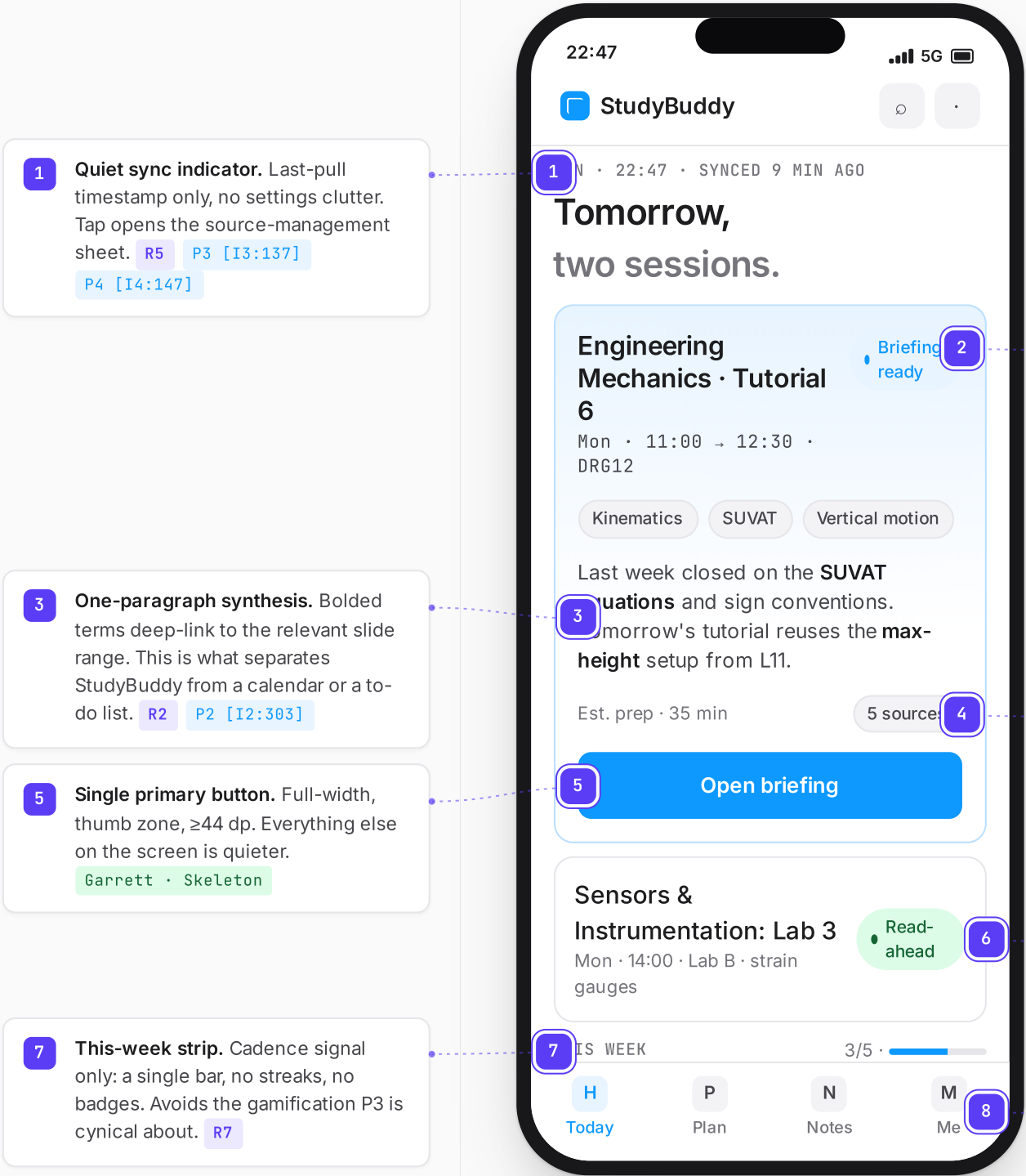
INTERACTION SPEC

TOUCH
The whole briefing card is one tap target (≥44 dp). Topic chip row caps at 3 by default; if more topics surface, the row scrolls horizontally with a fade edge.

TRANSITIONS
Card tap routes to Screen 02 (push, ~200 ms ease-out). Pull-down on the scroll container forces a Moodle re-pull; success returns a 1 s confirmation strip on the eyebrow timestamp.

ERRORS / OFFLINE
Last good briefing remains readable without network (offline-first). Sync failure shows an inline retry banner above the card; tap retries with exponential back-off (2 s, 8 s, 30 s).

ACCESSIBILITY
Focus order: greeting → briefing button → secondary card → tab bar. Briefing button SR-label names module + slot ("Open Engineering Mechanics Tutorial 6 briefing"). Eyebrow timestamp uses aria-live="polite" so synced-time updates are announced. Dynamic type up to ×1.5 supported; chip row truncates with ellipsis at the larger sizes.



1 **Quiet sync indicator.** Last-pull timestamp only, no settings clutter. Tap opens the source-management sheet. R5 P3 [I3:137] P4 [I4:147]

3 **One-paragraph synthesis.** Bolded terms deep-link to the relevant slide range. This is what separates StudyBuddy from a calendar or a to-do list. R2 P2 [I2:303]

5 **Single primary button.** Full-width, thumb zone, ≥44 dp. Everything else on the screen is quieter. Garrett · Skeleton

7 **This-week strip.** Cadence signal only: a single bar, no streaks, no badges. Avoids the gamification P3 is cynical about. R7

2 **Briefing card (the main action).** Module, slot and room from MyT timetable. Topic chips capped at three to avoid the "300 pages" intimidation P2 reported. R1 P2 [I2:295]

4 **Estimated prep time and sources count.** Time is the only plan-level number here, which supports P2's "60% ideal-session" realism. Norman · system image P2 [I2:257]

6 **Secondary card (next class).** Lower density. The "Read-ahead" chip mirrors P4's pre-lecture habit. P4 [I4:107] P1 [I1:81]

8 **Persistent tab bar.** Today / Plan / Notes / Me. Persistent destinations let the student recognise where to go rather than recall a route. Budiu · recognition

Home: tomorrow's briefing card

Single-card focus on the next study-relevant event, pulled from MyT timetable and ranked tomorrow-first. The card is the entry point to every other screen.

HCI PRINCIPLE APPLIED

Single primary action per screen. The briefing card itself is the primary action; the accent colour appears once, so the eye lands there first. Cooper · experience goal of control; Garrett · Surface for visual hierarchy.

EVIDENCE: INTERVIEW ANCHORS

Why a single tomorrow-card? P1 pre-reads next-day slides at their fixed evening block P1 [I1:81]; P4 reads ahead by topic before lectures P4 [I4:107]; P3 only structures the day when there is a project deadline P3 [I3:69]. The pre-class briefing formalises a behaviour **three of four already do manually**, and gives P3 the deadline-style scaffold they otherwise lack.

REQUIREMENTS TRACED

R1 R2 R5 R7

INTERACTION SPEC

TOUCH

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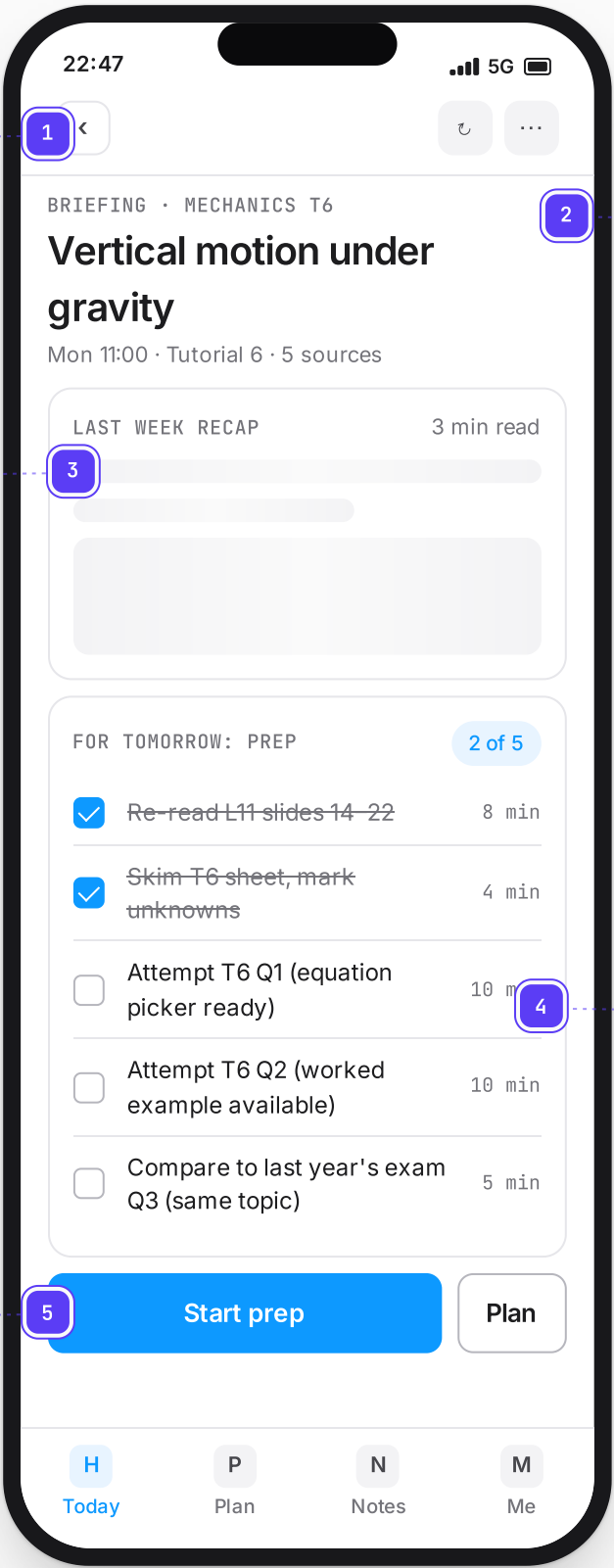
ACCESSIBILITY

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1 **Nav: refresh during load.** Refresh re-attempts the AI pull in case it stalled. Overflow exposes *use cached recap from yesterday* as the offline fallback. Cooper · experience goal

3 **Skeleton shape mirrors final content.** Three rows (full, short, tall) match the eventual recap row layout, so the student knows what is coming and where to look. No spinner: skeletons and the section header carry the progress signal.

5 **Primary button still routable.** "Start prep" jumps to the next incomplete item. Checklist items that depend on the recap mark themselves *"ready when recap loads"* and show a small notice when their dependency resolves.



2 **Header populates immediately.** Module, slot and sources-count come from MyTimetable and the local cache, with no AI synthesis required, so it never appears blank. The skeleton state begins below this line. Norman · system image

4 **Prep stays interactive.** Recap is the only AI-synthesised section; everything below remains tappable, so the student can start checklist items they already understand without waiting on the model.

Briefing detail: recap and prep

Mid-synthesis state. The header and prep section populate immediately from local data. The recap is the only part that defers on the AI pipeline, shown here as three skeleton rows mirroring the eventual layout. The student is never blocked waiting on the model.

HCI PRINCIPLE APPLIED

Skeleton screens, not spinners: showing the eventual shape keeps the student oriented (**Norman · system image**). The section header itself carries the progress signal; an extra spinner would widen the gulf of evaluation, not narrow it.

EVIDENCE: INTERVIEW ANCHORS

Why no spinner? P3's AI-cynicism P3 [I3:165] and P2's hallucination concern P2 [I2:169] both worsen with opaque waits. A spinner says "trust me, working" while a skeleton says "this is the shape you're getting". P2's ~60% ideal-session rate P2 [I2:257] means partial synthesis must yield partial value: even a half-loaded briefing should let the student start prep, so the screen unblocks the prep checklist while the recap is still resolving.

REQUIREMENTS TRACED

R2 R3 R4

INTERACTION SPEC

TOUCH
Source chips on each recap row are pill tap targets (≥ 40 dp). Prep checklist items accept a tap anywhere on the row to toggle. Refresh icon is debounced 800 ms to prevent rapid-fire pulls.

TRANSITIONS
Checklist tap shows a 3 s undo toast before commit (toast dismisses on tap or auto-collapses to a strikethrough). "Start prep" routes to the next incomplete item in order.

ERRORS
"Source missing" banner falls back to nearest available source and labels the substitution inline. Refresh failure preserves the previous recap and shows a one-line "Retry" link.

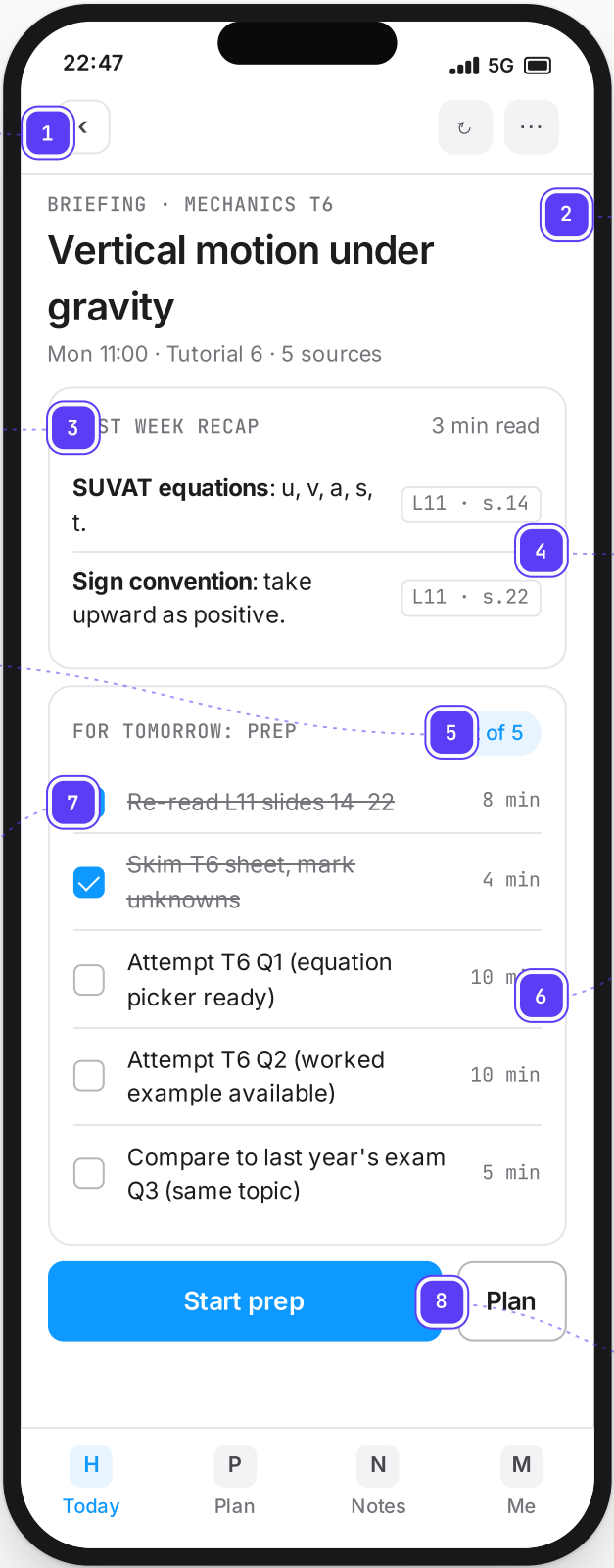
ACCESSIBILITY
Recap rows render as a `<dl>` with the source chip as `<dd>`. Checklist uses `role="checkbox"`; an `aria-live="polite"` region announces "n of 5 done" after each toggle. Reduce-motion replaces the strikethrough fill with an instant style change.

1 Nav: back, refresh, overflow. Refresh re-pulls Moodle in case slides have updated since the last sync. The overflow hosts *open in browser* and *report wrong source* for when the recap mis-attributes. Cooper · experience goal

3 Recap section: read-time. A soft commitment device ("3 min read") rather than a timer. Anchors P2's session-count goals without imposing one. P2 [I2:281]

5 Prep section: "n of m" progress chip. Live count of completed checklist items, so "ready" has an explicit definition the student can see at a glance.

7 Done items, two-tap undo. Tap triggers a 3s undo toast. Strikethrough on completion gives the visual reward; the undo window prevents accidental progress loss. Cooper · experience goal



2 Page header: what, when, source breadth. Module plus tomorrow's slot from MyTimetable, plus a "5 sources" line that shows where the briefing was built from *before* any synthesised text appears. R3

4 Recap rows: source on the row. Each topic links to its exact slide range (L11 · s.14). Attribution sits on the row, not buried in a footer, which is a direct answer to P2's hallucination concern and P3's AI-cynicism. R3 P2 [I2:169] P3 [I3:165]

6 Numbered, time-boxed checklist. Each item carries an estimate; items 3 and 4 deep-link into the equation picker (Screen 03) and worked solution (Screen 04). Item 5 routes the same flow at a past-paper question instead of a tutorial sheet, which is P3's go-to diagnostic and P4's exam-prep default. This is the entry point into T1. R4 P3 [I3:97] P4 [I4:301]

8 Two buttons (routing intent). *Start prep* (accent) routes to the next incomplete item, which is the default path. *Plan* (ghost) jumps to Screen 05 to adjust block sizing before starting, for nights where the proposed weighting feels wrong.

Briefing detail: recap and prep

Two stacked sections: a **recap** of last session (source-attributed) above **prep** for tomorrow (numbered, time-boxed, leading into the equation picker). Mirrors P4's last-session refresher ritual and gives P2 the lecture-to-tutorial link they currently have to construct manually.

■ HCI PRINCIPLE APPLIED

Progressive disclosure: the home card summarised in one paragraph; this screen is the drill-down. **Budiu · recognition vs recall:** every recap claim is a tappable source chip. **Cooper · experience goal:** refresh and undo are one tap each.

■ EVIDENCE: INTERVIEW ANCHORS

P4's *last-session refresher* ritual P4 [I4:259] shows students naturally want continuity from yesterday into today, but none of the four reported a tool that shows it automatically. P2's *"I'll try to find a solution inside the lecture notes, which never happens"* P2 [I2:311] is the gap this screen fills: the recap card converts a 20-minute scroll through Moodle into a 3-minute structured read. In exam-prep weeks the same screen accepts a past-paper question in place of a tutorial sheet, picking up P3's mock-paper diagnostic P3 [I3:97] and P4's exam-prep default P4 [I4:301] without a separate flow.

■ REQUIREMENTS TRACED

R2 R3 R4

■ INTERACTION SPEC

TOUCH
Source chips on each recap row are pill tap targets (≥40 dp). Prep checklist items accept a tap anywhere on the row to toggle. Refresh icon is debounced 800 ms to prevent rapid-fire pulls.

TRANSITIONS
Checklist tap shows a 3 s undo toast before commit (toast dismisses on tap or auto-collapses to a strikethrough). "Start prep" routes to the next incomplete item in order.

ERRORS
"Source missing" banner falls back to nearest available source and labels the substitution inline. Refresh failure preserves the previous recap and shows a one-line "Retry" link.

ACCESSIBILITY
Recap rows render as a <dl> with the source chip as <dd>. Checklist uses role="checkbox"; an aria-live="polite" region announces "n of 5 done" after each toggle. Reduce-motion replaces the strikethrough fill with an instant style change.

Tackles P2's headline finding head-on: equation selection is the b...

1 **Question header (verbatim).** No re-phrasing; trust requires the student see exactly what was asked. Tappable to expand to full PDF context. R3

3 **Section title with usage hint.** Long-press shows the derivation of the equation; tap selects. Two-level interaction keeps the primary surface clean. The candidate list is bounded at 2 to 4 deliberately: each extra option adds decision time, so we cap the visible set. Yablonski · Hick's Law

5 **Formula plus plain-language "why".** The "why" is the thing tutorials and slides skip, which is P2's "lecturer doesn't give you the procedure". P2 [I2:77, 157]

7 **Unlikely candidates: greyed but visible.** Shown, not hidden. Students learn what is wrong by seeing it dismissed with reasoning. Norman · mental model



2 **Diagram (extracted, not redrawn).** Real diagrams only. Gives the diagram equal weight with the question text, which is what P3 reaches for when picking an equation. P3 [I3:57]

4 **Selected: accent border and tint.** Numeric confidence ("92%") is shown **only** when high; a label otherwise. Avoids spurious precision. Norman · system image (calibrated)

6 **Support equation.** Surfaced alongside the primary so the alternative path (here: solve for t first, then s) is visible before any algebra starts. R4

8 **Show working / Pick another.** Routes into Screen 04 on accept; "Pick another" keeps the student in control if they disagree with the ranking. Cooper · experience goal

Equation picker: "which one?"

The screen that **refuses to give the answer**. For a tutorial question, StudyBuddy proposes 2 to 4 candidate equations ranked by relevance, with a one-line *why* and a confidence band. The student picks; only then does Screen 04 walk the working.

HCI PRINCIPLE APPLIED

Budiu · **recognition vs recall** and Yablonski · **Hick's Law**: the student picks from 2–4 ranked candidates rather than retrieving from memory; capped because decision time grows with choice count. Cooper · **experience goal**: "Pick another" at every step. Honest uncertainty: numeric confidence shown only when calibrated; otherwise a soft label.

EVIDENCE: INTERVIEW ANCHORS

This screen is the centrepiece of the design. P2's *"equation selection is the bottleneck"* P2 [I2:77] is the single largest finding across the four interviews. P3's diagram-to-equation mapping P3 [I3:57] and P4's "nearest tutorial" pattern-matching P4 [I4:313] describe the same workaround: manually scanning for a similar example because no tool offers candidate equations directly.

REQUIREMENTS TRACED

R3 R4 R6

INTERACTION SPEC

TOUCH
Whole eq-row is the selection target (≥56 dp tall). Long-press (500 ms) reveals derivation in a half-sheet that dismisses on backdrop tap. "Pick another" clears the current selection without leaving the screen; the previously-selected row remains visible.

TRANSITIONS
"Show working" → Screen 04 (push). Re-entering the picker preserves the prior selection and scroll position so the student can compare against the working without losing context.

ERRORS
"No match" empty state offers a manual "point to the right lecture" picker that re-runs ranking against the chosen module. Loading times above 1.5 s reveal a "scanning L9–L11..." footer.

ACCESSIBILITY
Eq-rows render as role="radiogroup"; the selected row has aria-checked="true". Confidence chip is decorative — the row's SR-label reads "high confidence", "support" or "unlikely" rather than a percentage. Dimmed "unlikely" rows remain in tab order and are still selectable; they are not aria-disabled.

1 Header: question ref, step count, source breadth. "Drawn from L11 and your textbook" names the sources before any working appears. [R3](#)

3 Working in monospace. Numbers, units and substitution are explicit. Final value bold. Directly addresses P1's gripe with Moodle's "560 kg" numerical-only solutions. [P1 \[I1:263\]](#)

5 "Why" line (one sentence). Plain-language rationale. Builds the mental model P4 wants. [P4 \[I4:309\]](#)

7 Override on AI rows. Lets the student replace the AI explanation with their own note for next time, which returns authorial control. [P3 \[I3:165\]](#)
[Cooper · experience goal](#)

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Q1 · WORKED SOLUTION

Maximum height of a vertical throw

3 steps · drawn from L11 + textbook

STEP 1 **Set up**[L11 · s.14](#)

Take upward as positive. $u = +12$ m/s, $a = -g = -9.81$ m/s², $v = 0$ at the peak.

Sign convention fixed before any algebra.

STEP 2 **Apply $v^2 = u^2 + 2as$** [L11 · s.16-18](#)

$0 = (12)^2 + 2 \cdot (-9.81) \cdot s$
 $19.62 \cdot s = 144$

$s = 144 / 19.62 = 7.34$ m

0 at the peak removes t from the equation.

STEP 3 **Solve for s**[Textbook · §12.5](#)

$s = 144 / 19.62 = 7.34$ m

Positive value confirms the upward sign convention.

Sources for this solution3 used

L11Engineering Mechanics, Lecture 11, slides 14–22open

TXTHibbeler · Engineering Mechanics: Dynamics, §12.5open

AIProcedure synthesis, narration onlyoverride

I've got it

Re-derive

2 Step block (uniform skeleton). Label, working, why, source. The same shape every step gives a quick visual scan, and addresses P3's view that missing intermediate steps means no learning. [P3 \[I3:121\]](#)

4 Source chip per step. Slide range, textbook section, or AI narration. Mixed-source steps get the most cautious chip ("AI · narration only"). [P2 \[I2:169\]](#)

6 Sources tray (separate, named, persistent). Source attribution lives in its own area rather than as a footnote. Lists every source and the role it played. [R3](#)

8 "I've got it" / "Re-derive". Felt-readiness is captured per question (not just per session), and feeds Screen 06 and tomorrow's weighting. [R7](#) [P3 \[I3:327\]](#)

Step-by-step worked solution

Numbered steps, each with a **label**, **working**(formula, values, result), a one-line **why**, and a **source chip**. A dedicated *Sources for this solution* tray sits at the bottom, surfacing attribution rather than burying it.

HCI PRINCIPLE APPLIED

Norman · mental model: symbols and ordering follow textbook convention so the system image matches what the student has built across L11 and the textbook. **Norman · 3 models:** the unverified-step state preserves the working but flags it, narrowing the gulf of evaluation when an AI synthesis cannot be pinned to a source.

EVIDENCE: INTERVIEW ANCHORS

Combined evidence: P1 [I1:263] numerical-only solutions; P2 [I2:157, 169] lecture skips procedure, AI-hallucination fear; P3 [I3:121, 165] intermediate-steps requirement, AI cynicism; P4 [I4:305, 309] worked example as mental model, lecturer-variable provision. **This screen alone hits all four participants on T1.**

REQUIREMENTS TRACED

[R3](#) [R4](#) [R6](#) [R7](#)

INTERACTION SPEC

TOUCH

Source chip tap opens the cited slide range in an external PDF reader (deep-linked to the slide number where supported, scroll-to-page otherwise). "I've got it" / "Re-derive" are the only screen-level buttons; in-tray actions (override) live inside the sources tray.

TRANSITIONS

Steps stream in sequentially as each resolves (200 ms stagger between cards). Reduce-motion replaces stagger with instant render. "Re-derive" re-requests synthesis, preserving the question and the previously-picked equation.

ERRORS

Unverified-step amber banner persists with the working still visible — never suppressed. "Switch source" deep-links into Settings → Sources. If all steps fail to verify, the screen falls back to "explain only" mode.

ACCESSIBILITY

Steps render as an ordered list. Working uses aria-label to expand symbols ("v squared equals u squared plus 2 a s"); the visual monospace formula is unchanged. The unverified-step banner is announced via role="alert" so SR users hear it before the step contents.

2 **Header unchanged.** Trust frame ("drawn from L11 and your textbook") is preserved so the student can see *which* source set was promised before comparing it to what was delivered. R3

4 **Unverified source chip (Step 3).** Warm tint, neutral copy ("unverified") rather than alarmist ("error" / "wrong"). Communicates uncertainty without implying the working is incorrect. Norman · system image (calibrated)

6 **"I've got it" still routes.** The student can still mark the question complete despite an unverified step; their felt-readiness rating (Screen~06) is what counts here, not the source tag. R7

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Q1 · WORKED SOLUTION

Maximum height of a vertical throw

3 steps · drawn from L11 + textbook

! Couldn't verify step 3 against L11. The procedure is shown but not source-attributed. Review carefully or switch source.

STEP 1 Set upL11 · s.14

Take upward as positive. $u = +12$ m/s, $a = -g = -9.81$ m/s², $v = 0$ at the peak.

Sign convention fixed before any algebra.

STEP 2 Apply $v^2 = u^2 + 2as$ L11 · s.16-18

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AIProcedure synthesis, narration onlyoverride

1 **Error banner (persistent).** Names the specific step that failed to verify and tells the student what to do next. Sits above the working so it is read first; does not dismiss on tap. P2 [I2:169] Norman · 3 models

3 **Working preserved, never suppressed.** P3 is explicit that intermediate steps are where the learning lives; hiding them on error would defeat the screen's purpose. The student sees the procedure and decides whether to trust it. P3 [I3:121]

5 **Switch source via Sources tray.** The override path lives in the same tray as the verified sources, so recovering from the error doesn't require leaving the screen. P3 [I3:165] Cooper · experience goal

Step-by-step worked solution

Error state. When the AI synthesis cannot be pinned to a slide range or textbook section, that step is flagged *unverified* rather than presented as authoritative. The working is preserved (never suppressed); the student can switch source, re-derive, or accept with reservation. This is the design's answer to P2's hallucination concern and P3's AI cynicism.

HCI PRINCIPLE APPLIED

Norman · 3 models: the system image must tell the student that its model of the answer is *incomplete*, not pretend otherwise. **Norman · gulf of evaluation:** the amber banner and the per-step chip show the uncertainty at the exact point it applies, so the student does not have to compare claims against sources by hand.

EVIDENCE: INTERVIEW ANCHORS

This is the design's answer to the AI-hallucination concern that recurred across the interviews: P2's "AI getting the calculation totally wrong --- showing you the right result, but not the right way to get there" P2 [I2:169] and P3's AI cynicism P3 [I3:165] both call for an explicit "don't trust this part" affordance rather than silent omission. Suppressing unverified steps would also breach P3's intermediate-steps requirement P3 [I3:121], so the working is preserved with the chip flagging it.

REQUIREMENTS TRACED

R3 R4 R6

INTERACTION SPEC

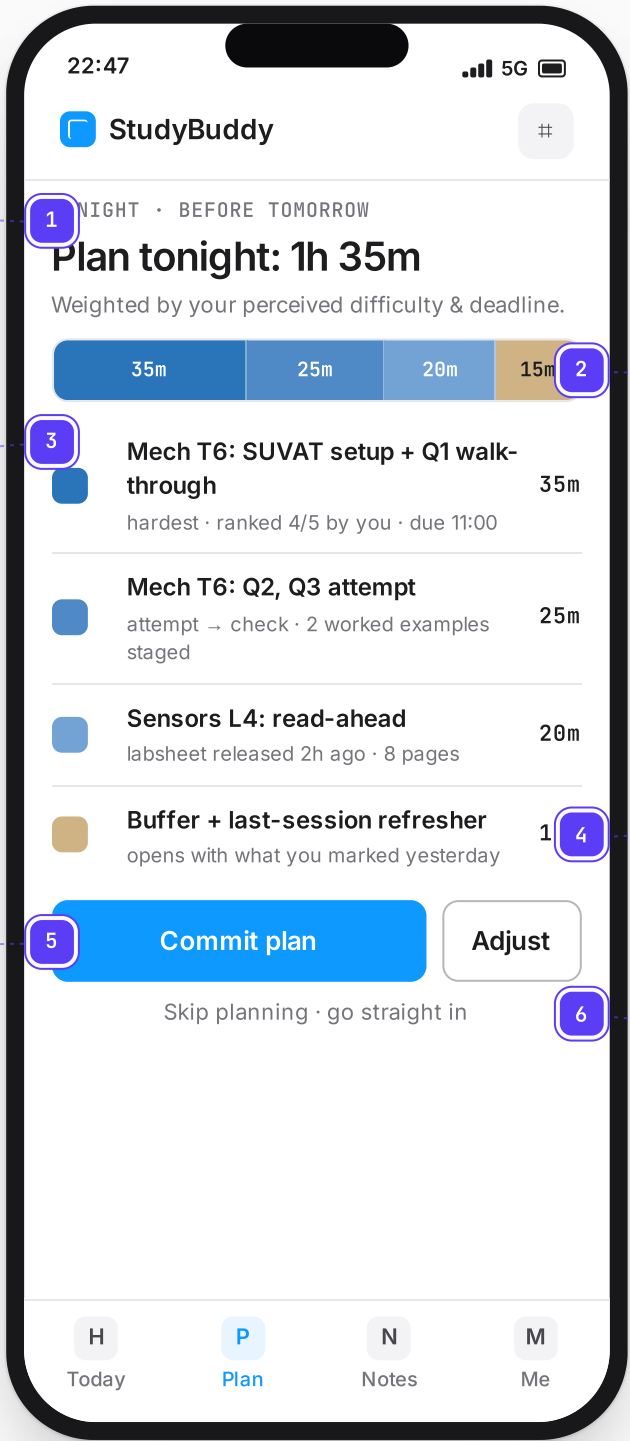
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Difficulty and deadline drive block sizes. Adjustable. Not a timer.



1 **Title: tonight, total estimate.** Time is the unit students already use mentally (P2's session-count goals). Deliberately avoids "studied this term" gamification. [P2 \[I2:281\]](#)

3 **Block list (one row per block).** Each row names *what*, *why this size*, and *when due*. The *why* is the bit students do not currently get from any tool. [R8](#)

5 **Commit / Adjust / Skip.** Three options at different commitment levels. *Commit* writes the plan into the home card; *Adjust* opens the inline editor; *Skip* bypasses planning entirely and routes back to Screen 02. [P3 \[I3:69\]](#)

2 **Plan-bar (proportional, four blocks).** Width is minutes; colour intensity tracks difficulty. Drag the divider between two blocks to reweight on the fly. [Shneiderman · direct manipulation](#)

4 **Last-session refresher block.** Built in; opens with what the student marked yesterday. Formalises P4's existing ritual. [P4 \[I4:259\]](#)

6 **No countdown, no in-session timer.** Deliberate omission. Focus is engineered environmentally (T3), not by app-imposed pressure.

Session plan / commit

A weighted plan-bar across four blocks, sized by perceived difficulty and deadline proximity. The student adjusts and *commits*, or skips planning entirely. There is deliberately no in-session timer (T3 is something the student arranges in their environment; the app shouldn't pile pressure on top of that).

HCI PRINCIPLE APPLIED

Garrett · Surface: one bar, four rows. No streaks, calendars or focus timers. **Shneiderman · direct manipulation:** dragging the divider reweights blocks in place, so the effect is visible on the surface that caused it.

EVIDENCE: INTERVIEW ANCHORS

Why difficulty-weighted, rather than time-blocked? P1 prioritises by difficulty P1 [I1:85]; P2 explicitly scales time to perceived difficulty P2 [I2:117]; P3 does the same P3 [I3:93]; P4 by motivation, which correlates with effort P4 [I4:127]. All four use difficulty (or its proxies) as the primary scheduling signal, not clock time.

REQUIREMENTS TRACED

[R7](#) [R8](#) [R9](#)

INTERACTION SPEC

TOUCH
Plan-bar dividers are draggable handles with a ≥ 24 dp hit area expanded from a 4 dp visual line. Per-row stepper appears in Adjust mode for ± 5 min increments. Total time is pinned at the top while dragging so the student sees the trade-off.

TRANSITIONS
Commit collapses the bar into a thin pinned strip on the home card and routes back to Screen 01. Skip routes to Screen 02 unchanged. Snooze hides the plan for 24 h with a "Snoozed" indicator on the home card.

ERRORS
"No prep needed" empty state replaces the bar with a snooze affordance. If the user drags two blocks to total under 5 min, the commit button disables with a tooltip ("Block too short").

ACCESSIBILITY
Plan-bar exposes one slider per block (role="slider", aria-valuemin / max / now in minutes, plus aria-valuetext for human-readable form). Drag is paired with stepper buttons for keyboard / SR users. Difficulty intensity is paired with text labels ("hardest", "medium", "lightest") — never colour-only.

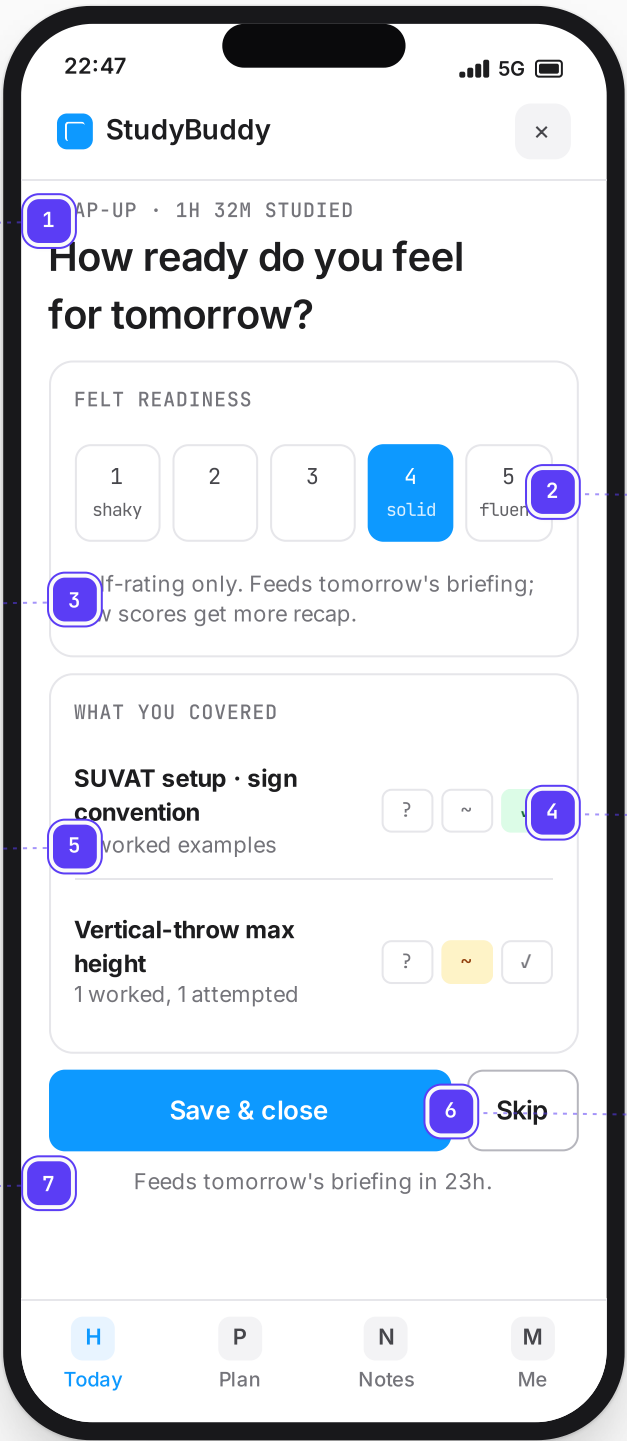
The loop. Capture what stuck and what didn't; feeds tomorrow.

1 **Header: minutes studied and the question.** "How ready do you feel?" is intentional: *felt*, not measured. Self-rating only — captures effort-as-success rather than output-as-success. P2 [I2:285]

3 **Soft explainer.** "Low scores get more recap" is transparent about how the data feeds back. Builds trust for P3's cynicism profile. P3 [I3:165]

5 **Coverage micro-copy.** "3 worked examples" or "read 6/8 pages" are observable, not inferred. Comes from logged interactions in Screens 03 and 04, not user input. Norman · system image

7 **Time-to-next-briefing footer.** "Feeds tomorrow's briefing in 23 h" closes the system-status loop the home screen opened.



2 **5-point readiness scale.** Endpoints labelled (*shaky* to *fluent*), so anchors are shared between sessions. Single tap; no confirm. ISO · efficiency

4 **Per-topic tally: ?/~/. Three states:** *didn't get to it*, *got partially*, *solid*. Captured per topic, not just per session, so the next briefing can be topic-specific. R7

6 **Save & close vs Skip.** Skip is a real option: wrap-up is voluntary. Skipped sessions still log time and items but do not influence next-briefing weighting.

Wrap-up · felt readiness

Three captures, in order: a 5-point **felt readiness** scale, a per-topic *?/~/. ✓* tally, and one optional free-text note. This screen is what makes the system more than a one-shot tool: it closes the loop into the next briefing.

HCI PRINCIPLE APPLIED

Norman · gulf of evaluation: the student tells the system whether the previous step worked, and the system *shows that input being used* on the next briefing. Without this loop, Screens 01–05 are unfalsifiable from the student's point of view.

EVIDENCE: INTERVIEW ANCHORS

P3's *"completion and felt-readiness signal"* P3 [I3:327] is the most explicit finding for this screen. P2's *"if it was tough and I worked a lot, I will say 'okay, that's fine for today'"* P2 [I2:285] shows effort, not output, as the success metric. It is captured here as the readiness scale, not a productivity score. P4's last-session refresher P4 [I4:259] needs **this** screen to populate the "what to refresh" list.

REQUIREMENTS TRACED

R7 R10

INTERACTION SPEC

TOUCH

Readiness scale is a single-tap-to-select segmented control (no drag); each segment is ≥44 dp wide. Outcome pills cycle through *?/~/. ✓* on tap. Both controls give a 30 ms haptic tick on select.

TRANSITIONS

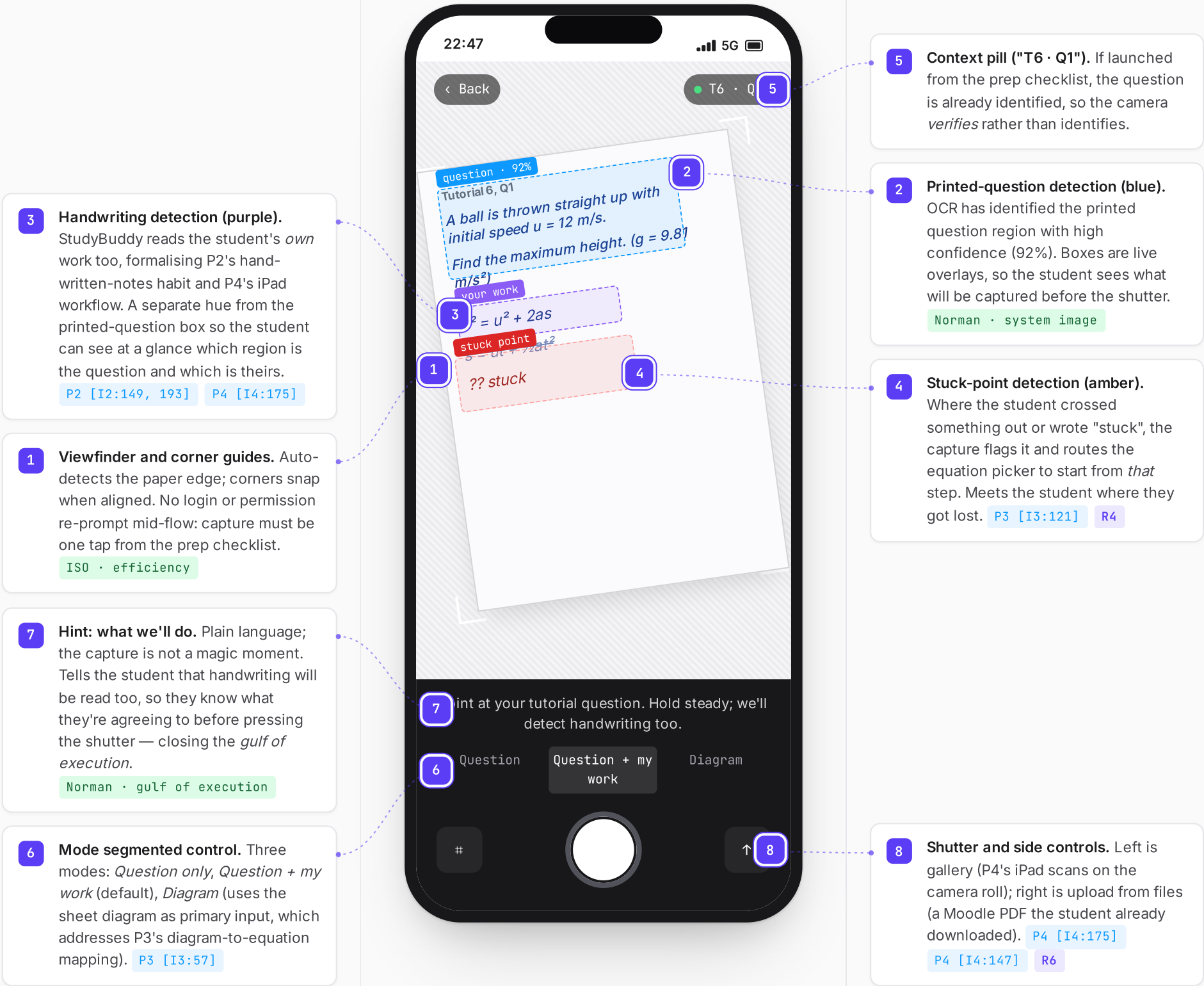
Save → 600 ms confirmation strip ("Saved, see you tomorrow") → auto-dismiss to Screen 01. Skip routes home immediately without the confirmation strip.

ERRORS

Save failure caches locally and retries on next launch; the user is not blocked. If three retries fail, a one-line banner on the home card surfaces "Wrap-up not yet synced".

ACCESSIBILITY

Scale renders as `role="radiogroup"` with endpoint labels ("shaky", "fluent") read by SRs regardless of selection. Pill states have text labels ("didn't get to it" / "got partially" / "solid"), so the symbol is decorative. Reduce-motion replaces the auto-dismiss animation with an instant route.



Camera capture: paper to briefing

Every participant studied on paper to some degree. This screen lets the student point their phone at a paper question (with or without their attempt) and routes into the equation picker pre-populated. Peer to "open from briefing", not a replacement.

■ HCI PRINCIPLE APPLIED

Norman · mental model: capture preserves the paper artefact rather than forcing retype. **Norman · gulf of execution:** the live OCR overlay and mode control tell the student what the camera will *do before* the shutter. Detection-box colour categorises what was found (blue for the printed question, purple for the student's own working, amber for stuck points); the shutter only commits what the boxes already show, and routing into Screen 03 means the AI never silently "answers" a captured question.

■ EVIDENCE: INTERVIEW ANCHORS

All four participants use paper or a tablet at some point: P1 attempts tutorials on paper before checking P1 [I1:255]; P2 prefers paper-style and hand-writes everything P2 [I2:149, 193]; P3 downloads tutorial PDFs and works at their desk P3 [I3:137, 169]; P4 uses an iPad with a smart pen P4 [I4:175]. A typed-input-only flow ignores the dominant medium.

■ REQUIREMENTS TRACED

[R3](#) [R4](#) [R6](#)

■ INTERACTION SPEC

TOUCH

Shutter is a 64 dp circular target. Mode segmented control switches the OCR pipeline live (no shutter press required to preview the new mode). Detection boxes are draggable on review (post-capture) with 8 dp corner handles.

TRANSITIONS

Successful capture freezes the frame and reveals review mode (300 ms cross-fade). "Use this" → Screen 03 with the captured question pre-populated. "Retake" returns to live preview without losing the previous mode selection.

ERRORS

Camera permission rejected: full-screen takeover with a guide to re-enable in OS settings, plus a "Use a photo from gallery" fallback. Low-light or paper-not-detected: shutter dimmed; corner guides pulse; hint replaced with corrective copy ("Too dark — turn on a light?" / "Frame the whole question and your working").

ACCESSIBILITY

Live OCR detection boxes are visual feedback only and not announced. A parallel aria-live="polite" summary announces the result on capture ("Question detected, your work detected, 1 stuck point detected"). Reduce-motion disables the box pulse and the cross-fade. Mode segmented control is a role="tablist" with text labels.